Inorganic Cumulative Exam Feb. 25, 2000

Chemical bonds can viewed as a continuum ranging from pure covalent to ionic bonding. Answer the following questions that deal with simple covalent and ionic bonding models for inorganic compounds.

1) The ΔH_f^o for an ionic compound can be *rationalized* in terms of a Born-Haber cycle. Discuss the reason for the formation of $MgF_2(s)$ and the absence of evidence for Mg(I) fluoride. Show the Born-Haber cycles for these two materials and indicate why the former has an exothermic ΔH_f^o and the the latter is anticipated to have an endothermic value for this thermodynamic quantity. *A quantitative solution is not required; however, a discuss of relative magnitudes of individual terms in the cycles is appropriate*.

2) The Pauling electronegativity (χ) is often used to assign the % ionic character in a chemical bond. Discuss the basis for this method of estimating ionic character and the relationship of this correlation to the assignment of Pauling electronegativity values.

3) Recently, the qualitative molecular orbital model has gained considerable popularity for discussing bonding in inorganic compounds. Show how the idea of a mixture of covalent and ionic character can be accommodated by this model using the diatomic NO.

4) The ligands NO and CO are predominately *N* and *C* donors respectively. Show that a simple MO model is consistent with these observations for one of these Lewis bases.